Persistent Homology of Music Network with Three Different Distances

POSTECH
MINDS
KIAS

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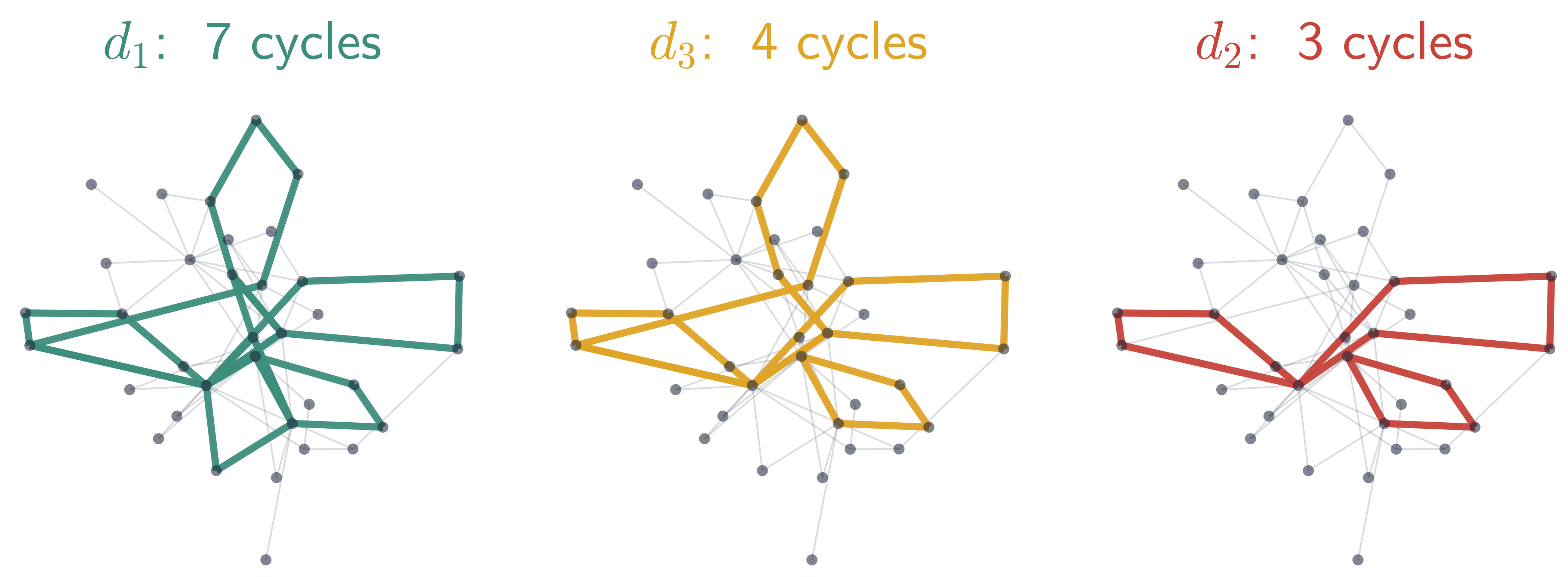
Project page

The big idea

Distance is a **tunable resolution knob** on musical structure — sweeping $d_1 \to d_3 \to d_2$ distills a piece down to its **core cyclic skeleton**.

- **Three path distances**: d_1 (min edges), d_2 (min weight — the only metric), d_3 (hybrid).
- **Proven ordering** $d_2 \le d_3 \le d_1$, lifting to a barcode injection $B_2 \subseteq B_3 \subseteq B_1$.
- **Validated** on real Korean court music — an exact match to the paper's Table 1.

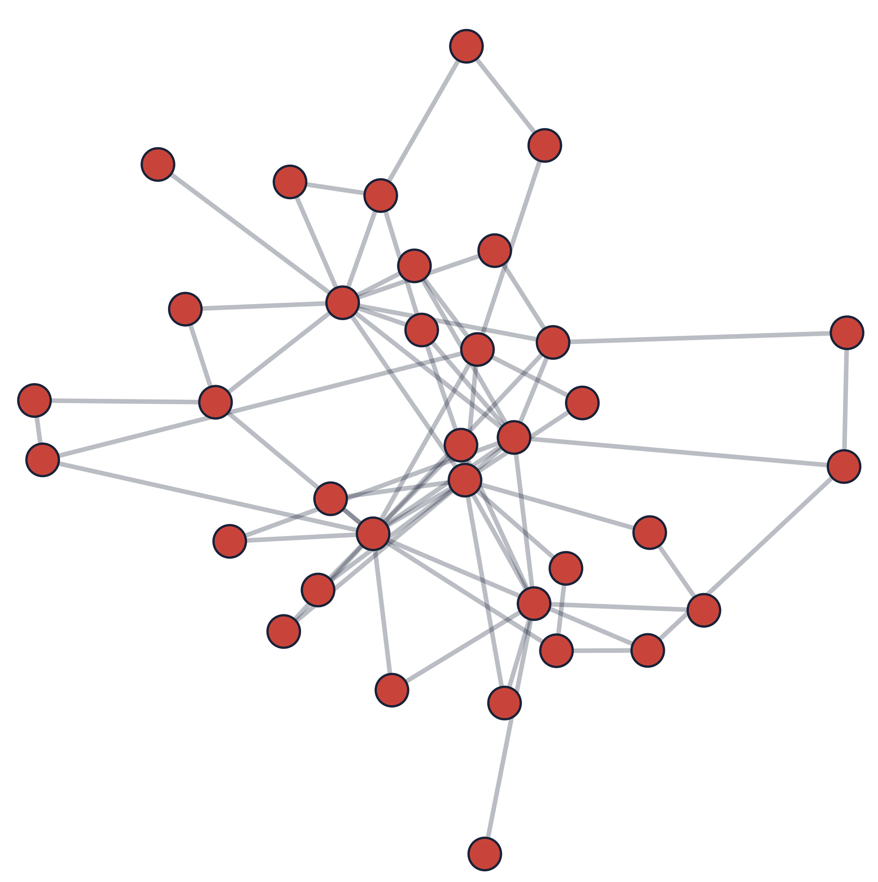
Cycle skeleton extraction



On *Sanghyeondodeuri* (geomungo): as $d_1 \to d_3 \to d_2$, cycles vanish $\mathbf{7} \to \mathbf{4} \to \mathbf{3}$ — only the robust core loops survive.

From music to a graph

Each **node** = (pitch, duration); an **edge** joins consecutive notes; the **weight** is 1/(co-occurrence count) — frequent neighbors are closer. Rests are skipped.



The music network of *Sanghyeondodeuri* (geomungo).

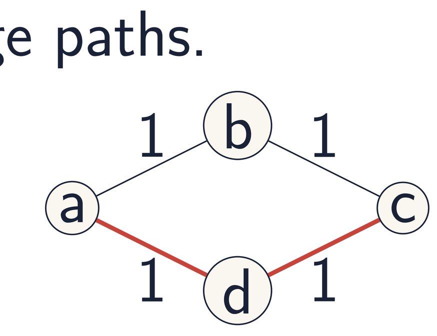
Three path distances

For nodes v, w on the music graph $M = (V, E, W)$:

d_1 **min-edge path** — fewest edges (weighted sum along it).

d_2 **min-weight path** — least total weight; *the only true metric.*

d_3 **hybrid** — min weight *among* the fewest-edge paths.



Path-based distance uses existing edges (red), not a new direct link.

Only d_2 is a metric

d_2 satisfies the triangle inequality; d_1 and d_3 can violate it (a heavy direct edge). Yet all three give meaningful musical structure.

A universal ordering

For *every* pair of nodes,

$$d_2(v,w) \le d_3(v,w) \le d_1(v,w).$$

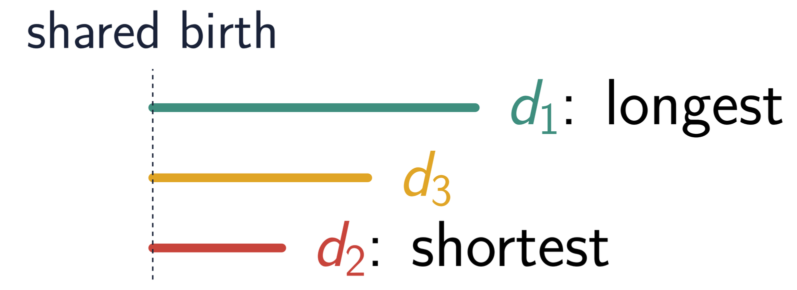
This carries over to one-dimensional persistent homology.

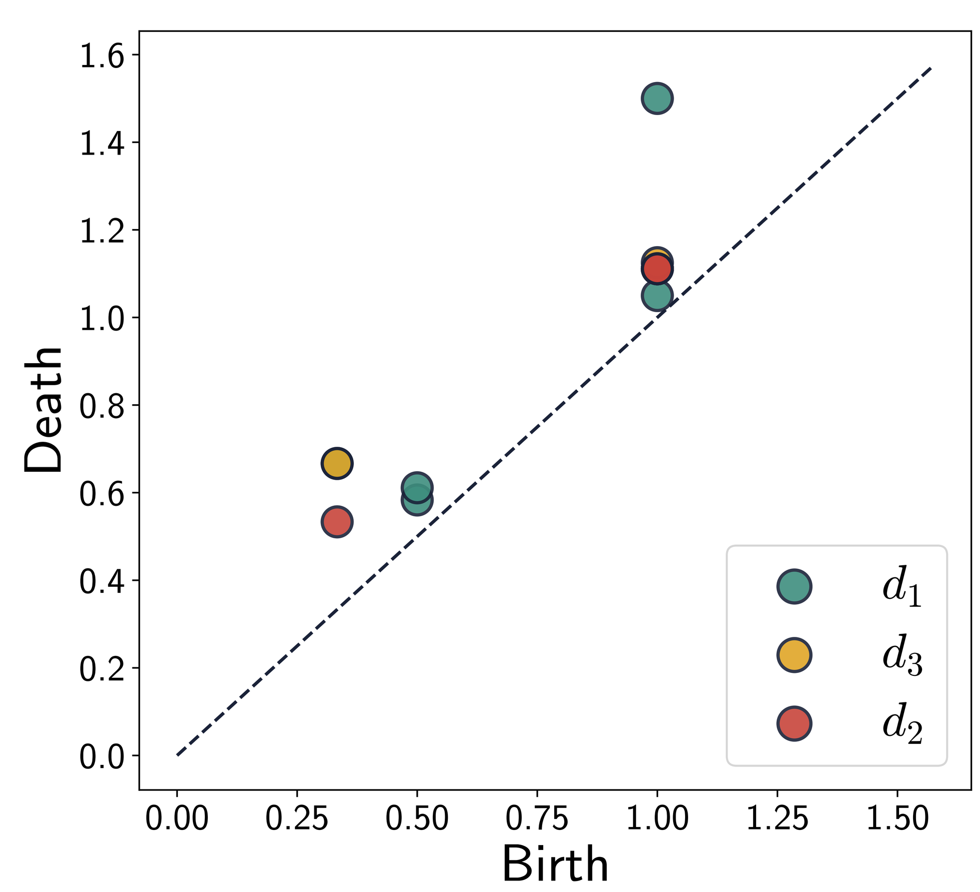
Barcode injection

The birth edges nest, $B_2 \subseteq B_3 \subseteq B_1$, giving injections

$$\mathrm{bcd}_1(d_2) \hookrightarrow \mathrm{bcd}_1(d_3) \hookrightarrow \mathrm{bcd}_1(d_1).$$

Each cycle keeps its **birth**; its **lifespan grows** $d_2 \le d_3 \le d_1$, and the number of cycles can only increase.



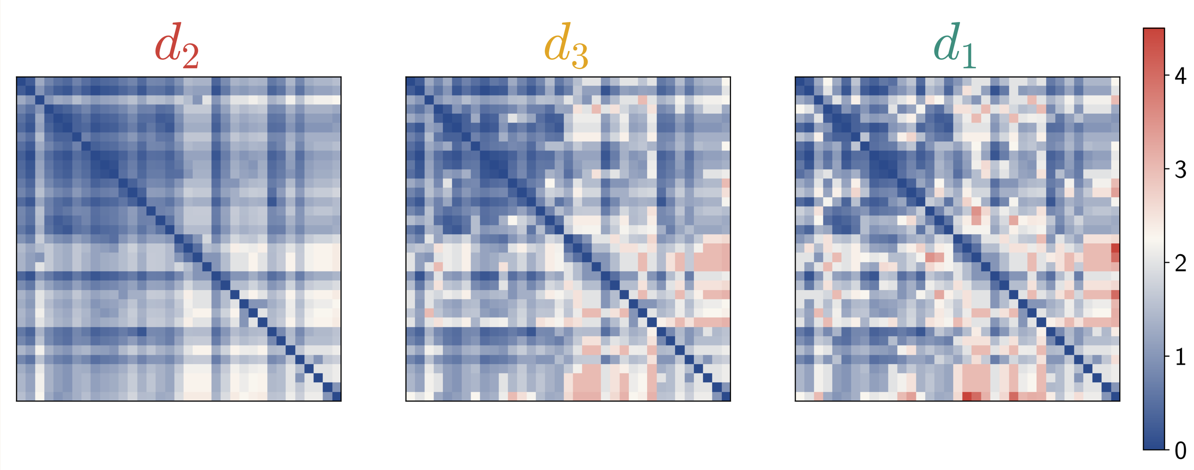


On real music: d_1 points lie above d_2 — longer lifespans, shared births.

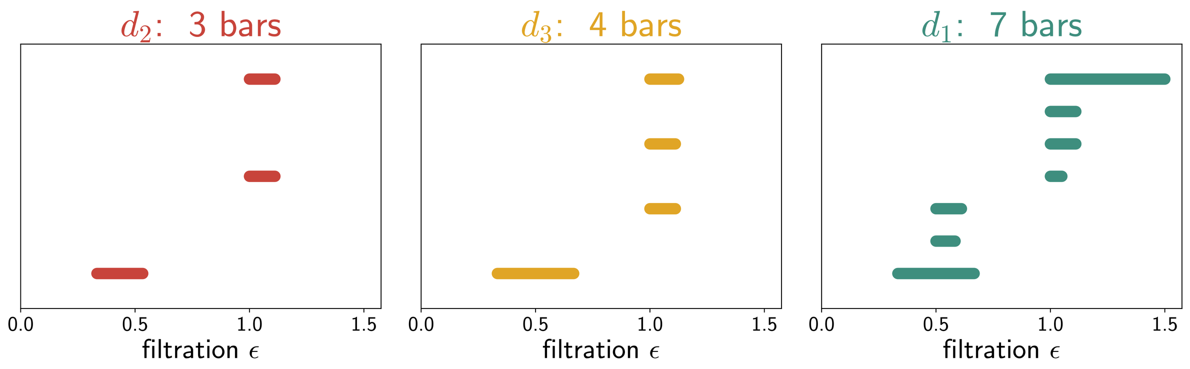
Why it matters

The same piece yields *different but nested* topological readings. d_2 exposes the robust core; d_1 adds finer, peripheral loops. The changes are **systematic, not random**.

Evidence on real music



Distance matrices: d_2 (cool/smooth) $\to$ d_1 (hot) — the ordering is visible.



H_1 barcodes: **7** / **4** / **3** bars for d_1/d_3/d_2; lifespans grow, births shared.

Data & validation

Twelve Korean court pieces (*geomungo* & *haegeum*) plus *Twinkle, Twinkle* variations. Our pipeline reproduces the paper's Table 1 **exactly** — same births, cycle counts, and inclusion relation.

Distilling the core

Ranking d_1 cycles by how far down they survive ($\to d_3 \to d_2$) gives a principled **importance hierarchy of musical loops** — a multi-resolution descriptor of a piece.

Takeaways & future work

- Distance choice = a tunable lens; results are nested, not arbitrary.
- Toward **AI music composition** with multiple stylistic perspectives from one source.

Interactive explorer & audio demos on the project page.

Key refs: Heo–Choi–Jung (2024) arXiv:2501.03553; Tran–Park–Jung (2023) *J. Math. Music*; Sethares–Budney (2014). Funded by NRF Korea (2021R1A2C3009648) & KIAS Transdisciplinary program. Contact: hew0920@postech.ac.kr This work: arXiv:2506.13595